

LIUZHOU, China

WMO: 590460

Lat: 24.3628N

Lon: 109.4600E

Elev: 1008

StdP: 14.17

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	37.7	40.2	21.7	16.9	50.6	26.2	20.9	48.9	14.4	43.5	12.3	46.2	5.5	0	0.320

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	95.4	78.4	93.9	78.3	92.5	78.0	81.0	90.2	80.3	89.2	79.7	88.3	5.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.7	154.8	85.1	78.1	151.5	84.7	77.5	148.3	84.3	45.7	91.0	45.0	89.5	44.3	88.5	84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.9	10.9	9.4	35.9	98.8	3.1	1.6	33.7	99.9	31.8	100.9	30.1	101.8	27.8	103.0		
			32.8	82.0	2.3	1.2	31.1	82.9	29.8	83.6	28.5	84.3	26.9	85.2		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	70.8	51.7	55.7	61.5	71.1	78.3	82.3	85.1	85.0	81.8	74.9	65.7	55.9	(6)
(7)		DBStd	13.31	7.75	9.29	7.76	7.25	5.00	3.91	3.45	3.54	4.89	5.99	6.75	6.37	(7)
(8)		HDD50	135	72	39	5	0	0	0	0	0	0	0	1	18	(8)
(9)		HDD65	1265	418	289	162	29	0	0	0	0	0	7	71	289	(9)
(10)		CDD50	7737	125	197	362	634	877	970	1088	1084	953	771	474	202	(10)
(11)		CDD65	3393	5	28	54	213	412	520	623	619	504	313	94	8	(11)
(12)		CDH74	35070	10	120	227	1411	3705	5712	7813	7681	5540	2366	461	24	(12)
(13)		CDH80	14375	0	19	31	342	1282	2255	3680	3622	2358	717	67	2	(13)
(14)	Wind	WSAvg	4.1	4.0	4.1	3.9	4.3	4.4	4.4	4.7	4.0	4.0	4.0	3.7	3.9	(14)
(15)	Precipitation	PrecAvg	59.7	2.5	1.9	4.0	5.2	9.7	12.9	7.5	6.5	3.0	2.3	2.4	1.9	(15)
(16)		PrecMax	90.5	7.4	5.2	12.6	12.0	15.2	24.8	16.3	20.6	10.1	7.8	10.3	6.4	(16)
(17)		PrecMin	37.9	0.2	0.1	0.2	1.3	4.8	4.6	2.1	1.3	0.2	0.1	0.1	0.2	(17)
(18)		PrecStd	12.2	1.6	1.2	2.3	2.5	2.7	5.8	4.2	4.0	2.2	2.0	2.3	1.6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	75.4	82.9	83.8	89.9	93.1	94.9	97.7	97.2	95.8	91.6	85.2	76.9	(19)
(20)		MCWB	64.8	69.1	71.4	74.9	77.4	78.6	78.9	77.9	77.1	73.9	69.7	67.0	(20)	
(21)		2%	DB	71.0	77.6	79.3	86.7	90.7	92.8	95.5	95.3	93.5	88.9	81.9	72.6	(21)
(22)		MCWB	61.8	66.5	69.4	74.0	76.4	78.7	78.9	78.2	77.1	72.8	68.2	61.9	(22)	
(23)		5%	DB	67.2	73.6	76.1	83.9	88.7	91.2	93.7	93.7	91.8	86.5	79.4	69.1	(23)
(24)		MCWB	59.1	65.3	67.8	73.1	75.9	78.3	78.8	78.3	76.6	71.9	67.0	58.5	(24)	
(25)		10%	DB	63.3	69.9	73.2	81.3	86.6	89.3	92.0	92.0	89.9	84.2	76.3	65.4	(25)
(26)		MCWB	56.1	63.0	66.1	72.2	75.5	78.1	78.5	78.2	76.0	70.5	65.5	55.4	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	66.6	70.2	73.1	77.4	80.7	81.5	81.9	81.8	80.1	77.6	74.0	70.5	(27)
(28)		MADB	73.0	80.2	80.8	85.6	89.3	90.5	92.1	90.8	89.4	86.5	79.4	75.4	(28)	
(29)		2%	WB	64.0	68.0	71.4	76.0	79.2	80.5	80.9	80.8	79.4	76.2	71.9	64.9	(29)
(30)		MADB	69.4	75.7	77.7	83.7	87.3	89.0	90.7	89.8	88.1	84.8	79.4	70.8	(30)	
(31)		5%	WB	60.6	66.0	69.7	74.7	78.1	79.8	80.1	80.1	78.7	74.8	70.2	60.5	(31)
(32)		MADB	66.0	72.7	74.9	81.7	85.7	87.7	89.3	88.9	87.0	83.0	77.3	65.9	(32)	
(33)		10%	WB	57.0	63.7	67.2	73.6	77.1	79.1	79.5	79.5	78.0	73.4	67.6	57.3	(33)
(34)		MADB	62.2	69.6	71.9	80.0	84.0	86.5	88.3	88.1	86.3	81.1	73.9	63.7	(34)	
(35)	Mean Daily Temperature Range	MDBR	9.5	9.8	9.6	10.5	11.3	10.3	11.3	11.6	12.2	12.3	12.3	11.4	(35)	
(36)		5% DB	MCDBR	17.4	16.4	15.2	14.6	14.8	12.9	13.8	13.9	14.1	15.0	16.3	17.7	(36)
(37)		MCWBR	8.9	7.9	7.4	5.7	4.8	3.6	3.0	3.2	3.5	4.8	6.3	8.5	(37)	
(38)		5% WB	MCDBR	14.2	14.5	12.6	12.4	11.7	10.7	11.8	12.3	12.6	12.9	12.4	13.6	(38)
(39)		MCWBR	8.8	8.3	7.7	6.5	5.5	4.1	3.7	3.9	3.9	5.4	6.4	8.6	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.623	0.652	0.775	0.741	0.599	0.567	0.534	0.604	0.638	0.695	0.597	0.616	(40)	
(41)		taud	1.622	1.613	1.430	1.534	1.868	1.970	2.059	1.850	1.743	1.579	1.730	1.661	(41)	
(42)		Ebn at Noon	199	206	190	202	232	238	246	229	217	193	204	193	(42)	
(43)		Edh at Noon	74	79	100	91	65	59	54	66	72	81	66	69	(43)	
(44)	All-Sky Solar Radiation	RadAvg	607	685	780	1008	1232	1253	1539	1514	1401	1184	906	731	(44)	
(45)		RadStd	152	146	120	129	128	170	140	101	109	115	154	148	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	-1.94	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	N/A	N/A	N/A	+0.76	N/A	N/A	N/A	N/A	N/A	+81

Nomenclature: See separate page